Charging at the bend of a finger

Chinese and US researchers have developed a small metallic tab that, when attached to the human body, generates electricity from bending a finger and other simple movements. So, searching for a power outlet may soon become a thing of the past. Instead, devices will receive electricity from the tab—a triboelectric nanogenerator.

The researchers' study, published online in the journal *Nano Energy*, describes the tab as being 1.5x1cm^2 in size. It delivered maximum voltage of 124 volts, maximum current of 10 microamps and maximum power density of 0.22 milliwatt per square centimetre. That is not enough to quickly charge a smartphone, however, it lit 48 red LED lights simultaneously.

Triboelectric charging occurs when certain materials become electrically charged after coming into contact with a different material. Most everyday static electricity is triboelectric, according to the collaborative research led by University at Buffalo and Institute of Semiconductors at Chinese Academy of Sciences.

A prototype triboelectric nanogenerator that can generate electricity from the bending of a finger (Credit: Nano Energy)

The tab consists of two thin layers of gold, with polydimethylsiloxane (PDMS), a silicon-based polymer used in contact lenses, Silly Putty and other products, sandwiched in between. Stretching one layer of gold causes it to crumple upon release and create what looks like a miniature mountain range. When that force is reapplied, for example, from a finger bending, the motion leads to friction between the gold layers and PDMS. This causes electrons to flow back and forth between the gold layers. The more the friction, the greater is the amount of power produced.

Sprint gearing to roll out the world's first mobile 5G network

Following a series of 5G rollout announcements from AT&T, Verizon and T-Mobile, US carrier Sprint has committed to launching mobile 5G service nationally in the first half of 2019—at somewhat higher prices. Unlike Verizon, which thus far has announced only 'fixed 5G' plans for wireless residential and commercial broadband service, Sprint is focusing on 'mobile 5G' portable wireless devices. It is working with Qualcomm on 5G technology, and a Korean manufacturer to ready 5G handsets for its network.

Unlike rivals that plan to utilise low-frequency and ultra-high-frequency spectrum in building 5G infrastructure, Sprint

Sprint CEO Marcelo Claure has claimed that Sprint will have the first mobile 5G network in the US and "potentially in the world" (Image credit: Reuters/ Mike Blake)